The `mtcars` data frame is included in your R session without needing to load a package. It includes several measures on different types of cars. Learn more about the data set using `?mtcars`.

## Question 1

We would like to study the relationship between cars' fuel efficiency and their weight per gallon.

### part a

Summarize the association between the fuel efficiency (measured in miles per gallon) and the weight of the car by writing code to create a scatter plot. Copy the code you used here.

### part b

Calculate the value of the correlation coefficient between fuel efficiency and weight below using `dplyr` code. Copy the code below, as well as the value of the correlation coefficient.

## Question 2

### part a

Fit a linear model using `lm` to estimate fuel efficiency using weight. Write the code you used below.

### part b

Based on the model output, write down the mathematical equation of the linear model.

## Question 3

### part a

Repeat **Question 1, part a** but use the horsepower of the car instead of the weight as the explanatory variable. You do not need to copy the code below.

**part b**

Calculate the value of the correlation coefficient between fuel efficiency and horsepower below using `dplyr` code. Copy the code below, as well as the value of the correlation coefficient.

**part c**

Repeat **Question 2 (parts a and b)**. You do not need to copy your code, but do write down the mathematical form of the linear model used to explain fuel efficiency with horsepower.

## Question 4

**part a**

What is the better way to compare the strength of the linear relationship between these two pairs of variables (mpg and wt; mpg and hp): the correlation coefficients or the slopes of the linear models?

**part b**

Explain your reasoning in one to two sentences.

**Question 5**

Out of all cars in the data frame, which car has the most surprisingly low fuel efficiency that we would expect to see given its weight (in other words, what is the car with the *most negative* residual value)? State the car and provide supporting code for your answer.